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INTRODUCTION TO INTERNET THE THINGS

Unit-1:Internet of Things Overview

SEMESTER: 5

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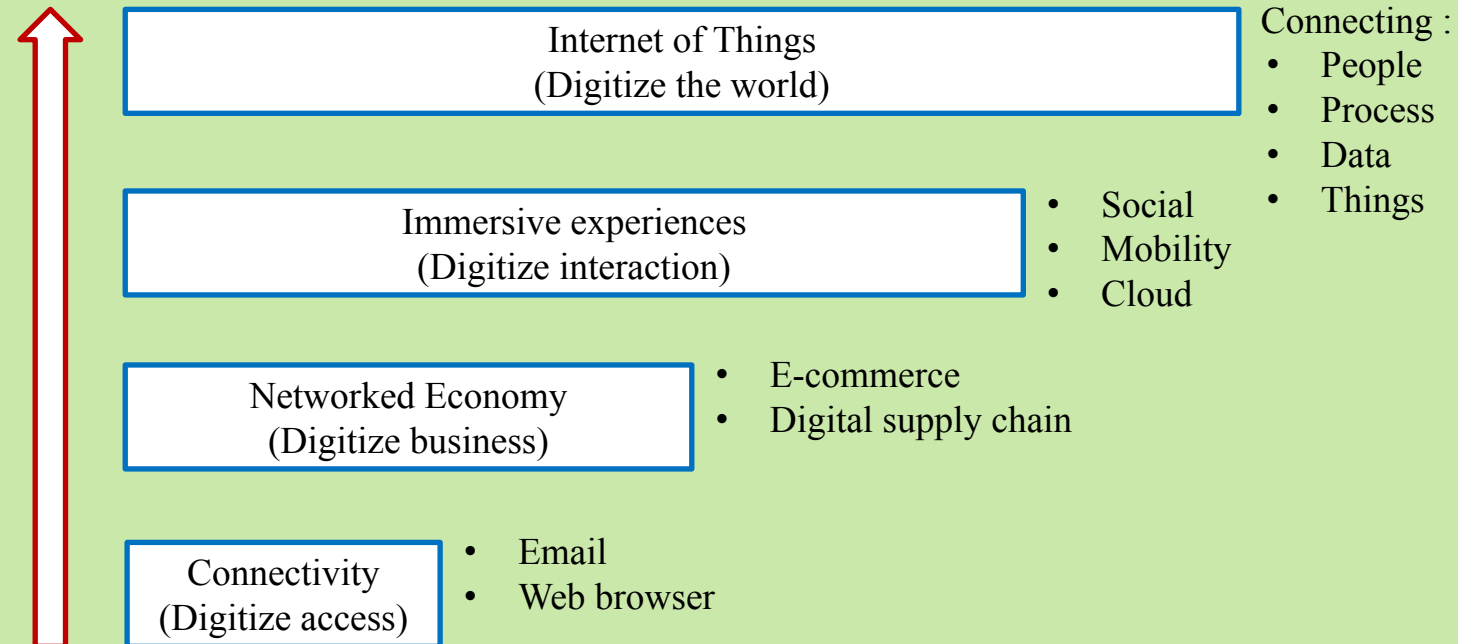
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LEARNING OBJECTIVES

1. Understand the evolution, definition, and characteristics of IoT in modern technology.
2. Learn the key components and architecture of IoT systems, including M2M and end-to-end models.
3. Analyze the advantages, disadvantages, and Interdependencies of IoT with cloud computing.

Evolution of IoT

Evolutionary phase of internet is connectivity, networked economy, immersive experience and IoT.



DEFINITION OF INTERNET OF THINGS

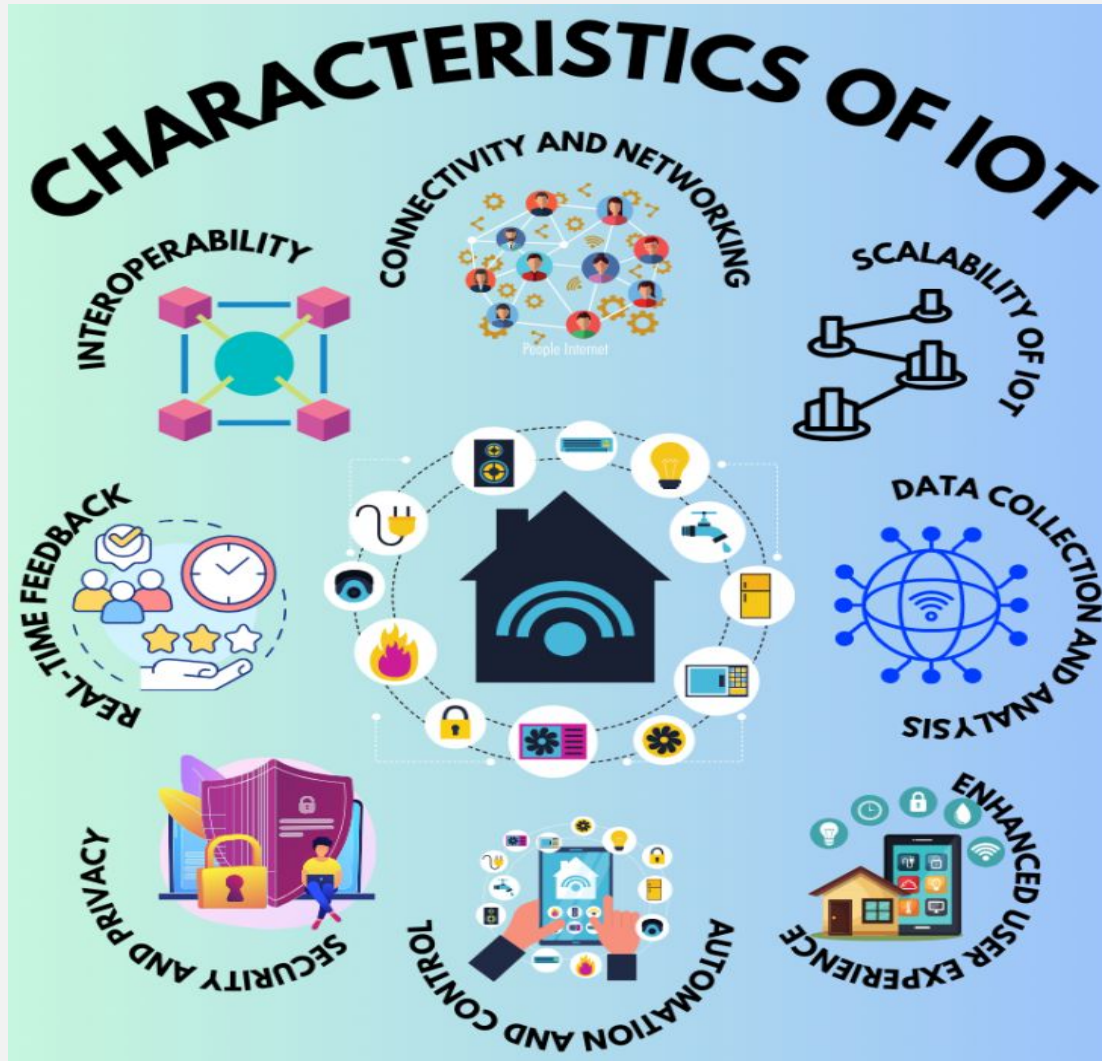
There are multiple ways to define IoT, but the basic of all the definitions remains the same.

IoT is network of interconnected computing devices which are embedded in everyday objects, enabling them to send and receive data.

OR

A dynamic global n/w infrastructure with self configuring capabilities based on standard and interoperable communication protocols where physical and virtual —things|| have identities, physical attributes and virtual personalities and use intelligent interfaces, and are seamlessly integrated into information n/w, often communicate data associated with users and their environments.

CHARACTERISTICS OF IOT



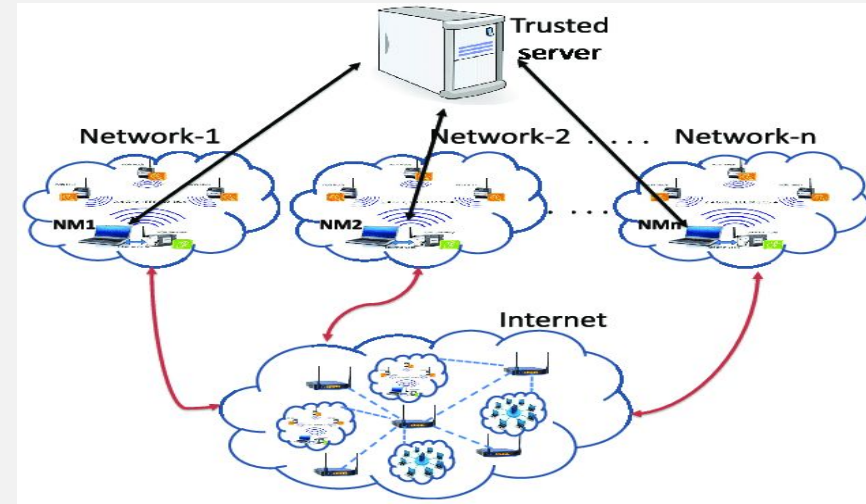
The Internet of Things (IoT) is characterized by the following key features.

- Connectivity & Networking
- Scalability
- Dynamic and Self-Adapting (Complexity)
- Architecture
- Safety
- Self Configuring
- Real-time Feedback
- Interoperability
- Identity of Things
- Automation & Control

CHARACTERISTICS OF IOT

Connectivity & Networking

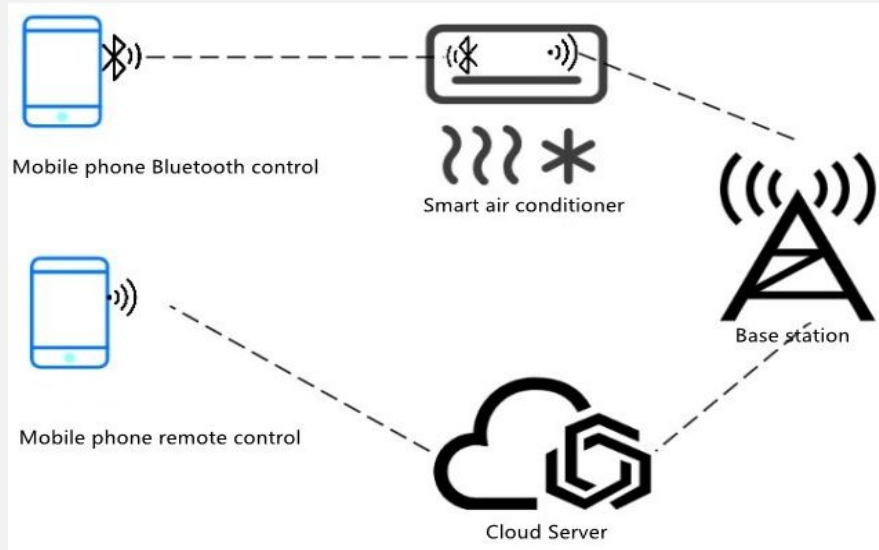
- Connectivity is an important requirement of the IoT infrastructure.
- Things of IoT should be connected to the IoT infrastructure.
- Anyone, anywhere, anytime can connect, this should be guaranteed at all times.



Scalability

- The number of elements connected to the IoT zone is increasing day by day.
- The IoT setup should be capable of handling the massive expansion.
- It can be either expand capability in terms of processing power, Storage, etc as vertical scaling or horizontal scaling by multiplying with easy cloning.

CHARACTERISTICS OF IOT

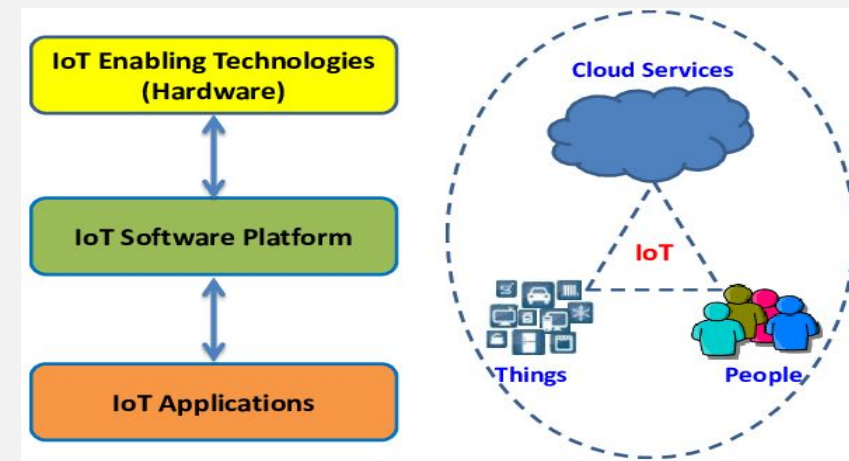


Dynamic and Self-Adapting (Complexity)

- This is achieved through dynamic changes that occur in the environment around these devices.
- an AC should have a sensor that can send a signal to the cloud and adjust it to the premises of the place.
- Similarly, the camera can easily click photographs by adjusting to light situations, like day and night.

Architecture

- The architecture of IoT is designed in a way that it is capable of supporting various devices, technologies, and protocols.
- it is essential for the devices to support diverse technologies and protocols and communicate with each other.



CHARACTERISTICS OF IOT

Safety

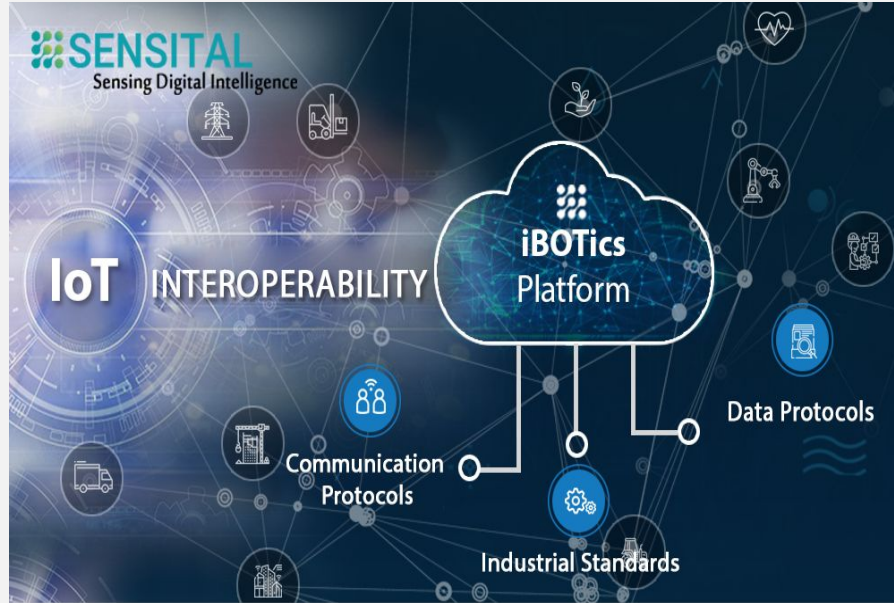
- when things are connected to the internet in such a way there is always a danger of the sensitive personal details of the users getting compromised.
- So safety is undoubtedly a crucial characteristic of IoT.



Self Configuring

- IoT devices are able to upgrade their software in accordance with requirements with a minimum of user participation.
- Additionally, they can set up the network, allowing for the addition of new devices to an already-existing network.

CHARACTERISTICS OF IOT



Interoperability

- Interoperability is critical for the success of IoT, as it enables different devices and systems to work together seamlessly and provides a seamless user experience.
- Without interoperability, IoT systems would be limited to individual silos of data and devices, making it difficult to share information and create new services and applications.

Real-time Feedback

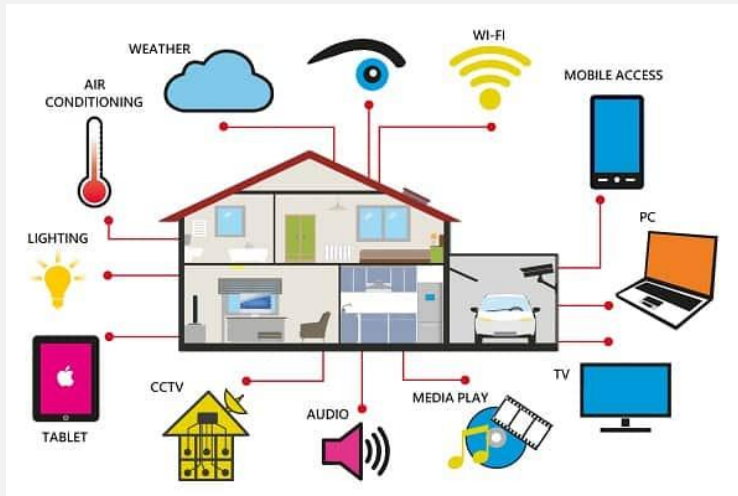
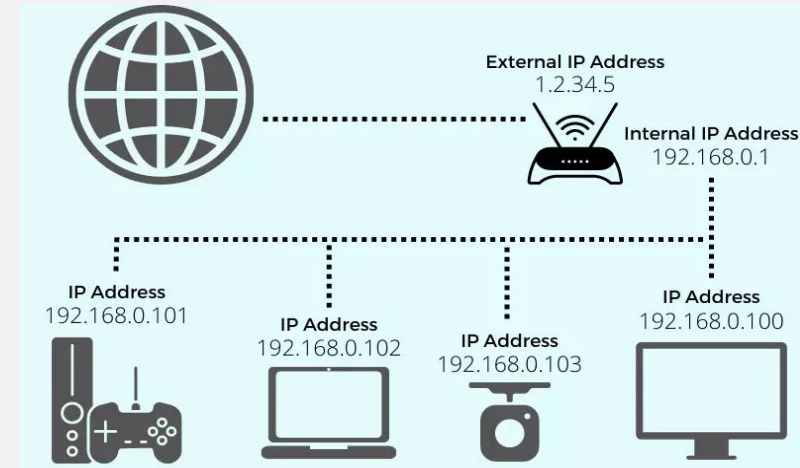
- Real-time feedback is a fundamental characteristic of IoT systems, enabling them to sense, analyze, and act upon data in real-time to achieve desired outcomes efficiently and effectively.



CHARACTERISTICS OF IOT

Identity of Things

- Each IoT device has a unique identity (e.g., an IP address).
- This identity is helpful in communication, tracking and to know status of the things.
- We can set up passwords or other security means.



Automation & Control

- automation and control are key characteristics of IoT systems.
- Enabling them to streamline processes, optimize resource utilization, and improve decision-making through remote operation, feedback loops, rule-based logic, and advanced analytics.

COMPONENTS OF IOT SYSTEM

The hardware involved in an IoT system includes remote dashboard devices, control devices, servers, routing or bridge devices, and sensors. These devices manage key tasks and functions, such as systematization, action specification, security, messaging, and pass-specific target and action detection.

Sensors or devices:

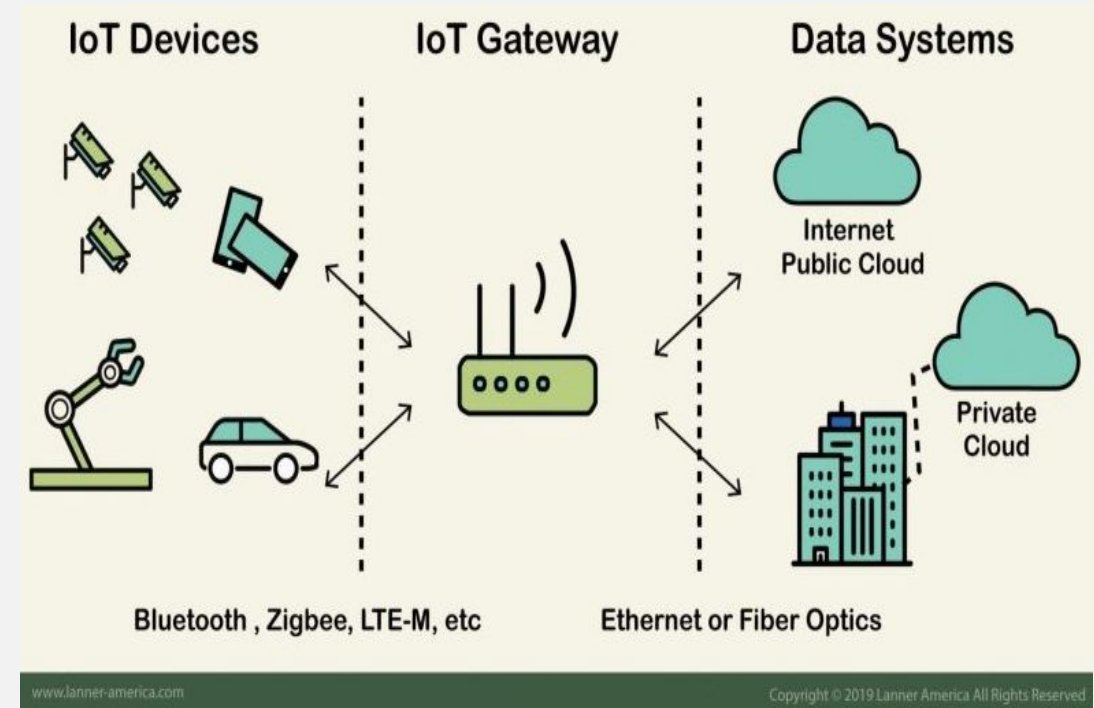
- Basically sensors or devices are used to collect and transmit data and perform actions based on that data. For example, temperature and humidity can be measured.
- Here are the different types of sensors: temperature sensors, humidity sensors, proximity sensors, motion sensors, light sensors, pressure sensors, gas sensors and GPS sensors.



COMPONENTS OF IOT SYSTEM

Gateway:

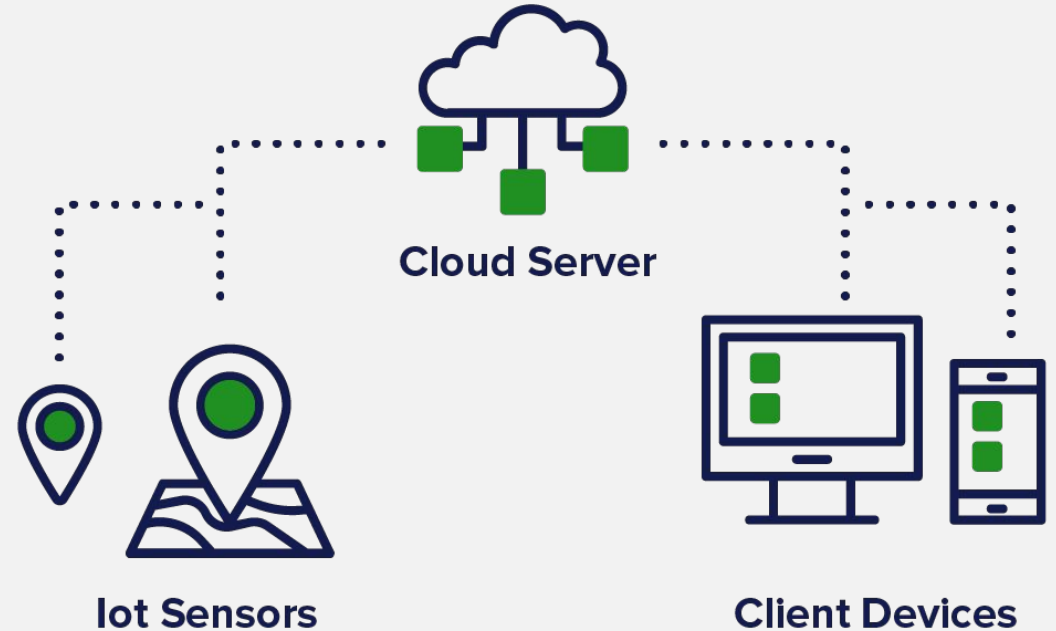
- A gateway is a device that basically acts as an intermediary between the sensor and the central cloud.
- Gateways are one of the essential components of IoT to provide communication, management and data processing.
- Gateways in IoT have several functions such as data aggregation, communication, security, protocol translation, load balancing and latency reduction.



COMPONENTS OF IOT SYSTEM

Cloud

- Cloud in IoT refers to a service that provides management, storage and processing of data generated by IoT (Internet of Things) devices.
- IoT has some key aspects of cloud like: data storage, data collection, security, connectivity, integration and cost efficiency.



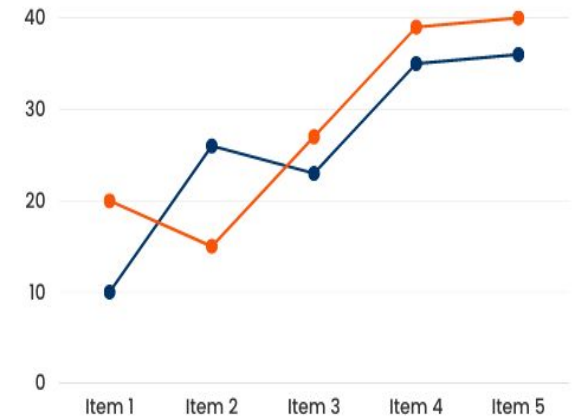
COMPONENTS OF IOT SYSTEM

Analytics

- This is an important and critical part of IoT that makes the most of its potential by analyzing useful information from IoT devices and sensors.
- Analytics involves tasks like data processing, machine learning, and statistical analysis. IoT uses analytics in many ways, such as finding unusual patterns, monitoring the environment, managing energy, and supporting smart cities and farming.

IoT Analytics

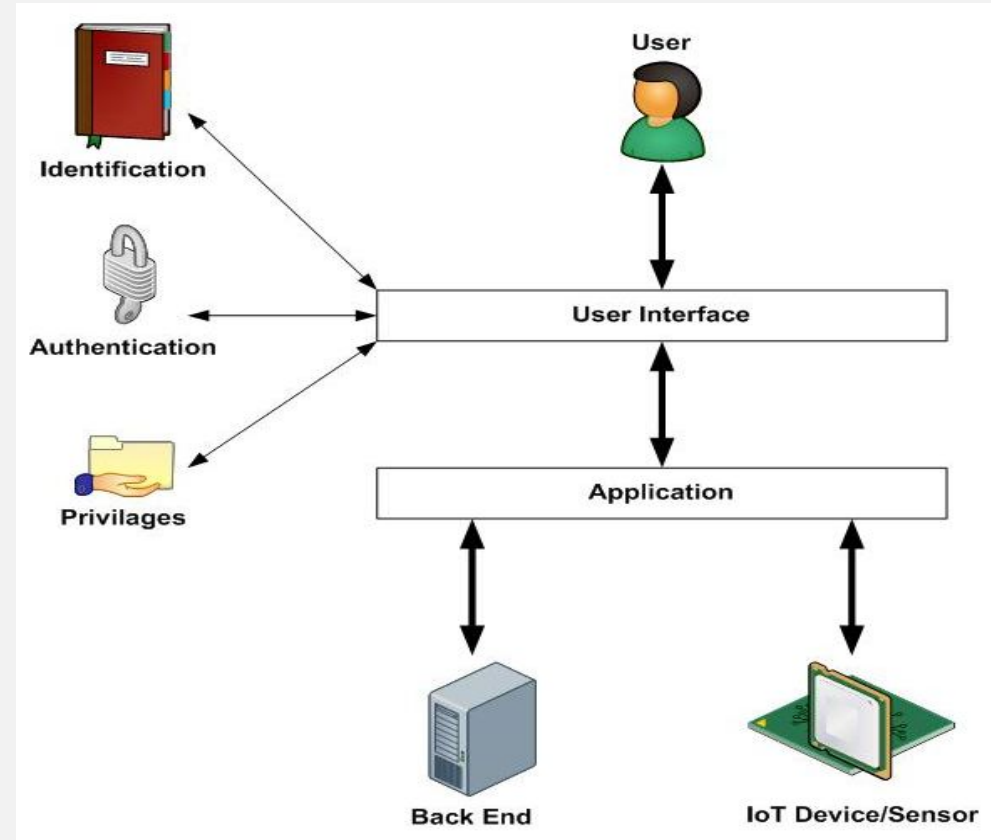
- Improved customer experience
- Increased efficiency
- Improved decision-making
- New business opportunities
- Streamlined operations
- Scalability



COMPONENTS OF IOT SYSTEM

User Interface

- User end application where user can monitor or control the data.
- A user interface, also known as a UI in the Internet of Things (IoT), provides an interface.
- Users can interact with applications and systems through interfaces.
- Key features of IoT user interfaces include data visualization, easy-to-use design, personalization, remote control, integration, authentication, and security.



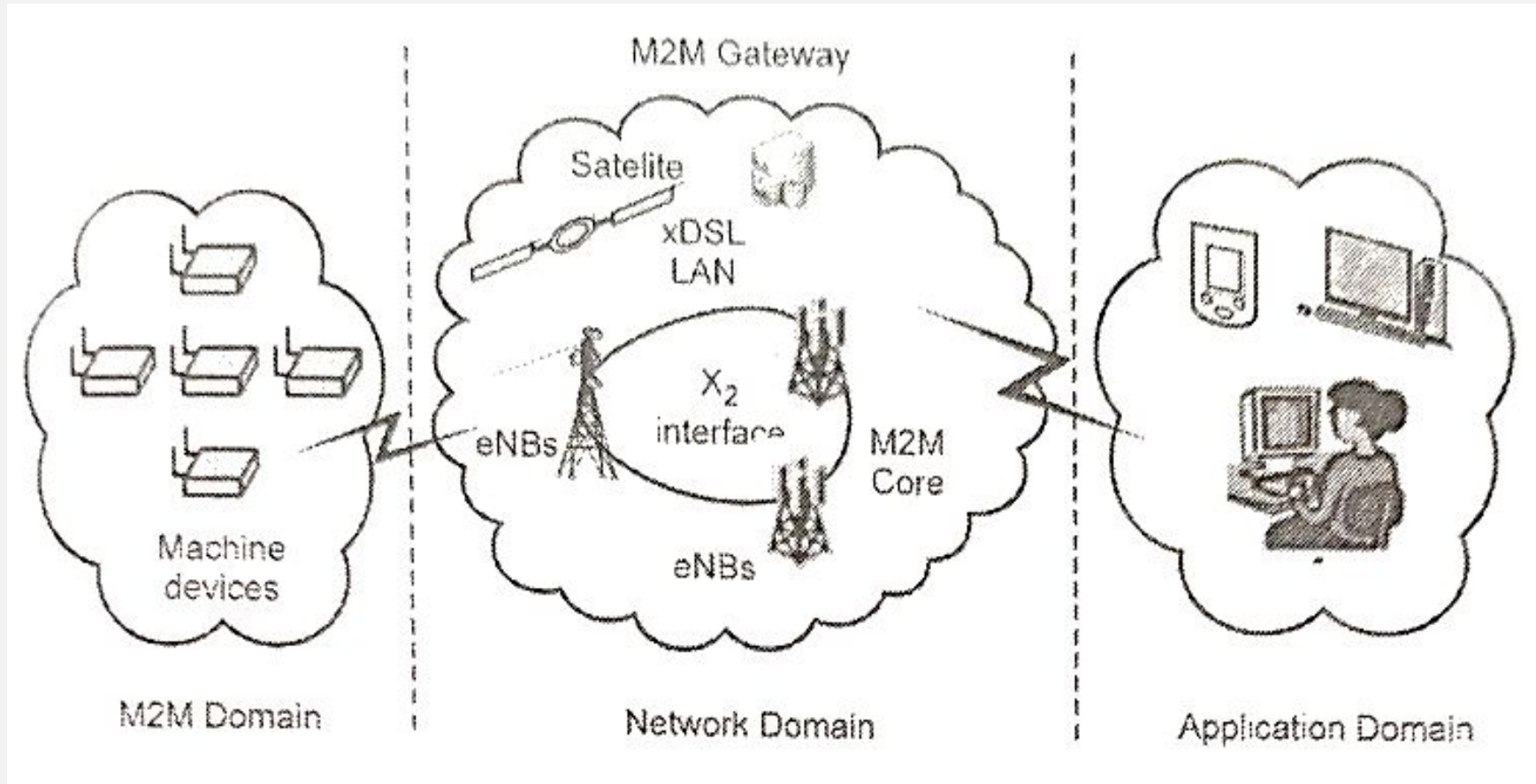
FUNCTIONING OF IOT SYSTEM

- The main parts of the Internet of Things (IoT) are devices that collect data. These devices can be tiny sensors, fitness trackers, or large machines doing complex tasks. Each device has a unique IP address because they are connected to the internet. The data they collect is sent to central places like cloud servers, where it is processed and analyzed so businesses can make decisions based on the results.
- Sometimes, real-time action is needed, making it inconvenient to send data to a central server. In these cases, edge computing is used. This means processing the data close to where it is collected, with the option to send it to a central location later if necessary.

M2M ARCHITECTURE

- **Definition:** M2M architecture focuses on the direct communication between devices or machines without human intervention.
- **Characteristics:**
 1. **Decentralized:** Devices communicate directly with each other or through an intermediary such as a gateway.
 2. **Device-centric:** Communication is primarily between devices, often following a peer-to-peer model.
 3. **Scalability:** Well-suited for large-scale deployments where numerous devices need to interact autonomously.

M2M ARCHITECTURE



- **M2M systems** typically focus on direct device communication for specific tasks like monitoring and control, often within constrained environments.

Fig.: M2M Architecture

FEW EXAMPLES OF M2M ARCHITECTURE:

1. Smart Agriculture:

Monitoring systems in agriculture use M2M communication between soil moisture sensors, weather stations, and irrigation systems. Sensors collect data autonomously and trigger actions like watering based on preset conditions, without human intervention.

2. Fleet Management:

GPS tracking devices in vehicles communicate directly with centralized servers or fleet management systems. This allows real-time monitoring of vehicle location, speed, and maintenance needs, enabling efficient fleet operations without constant human oversight.

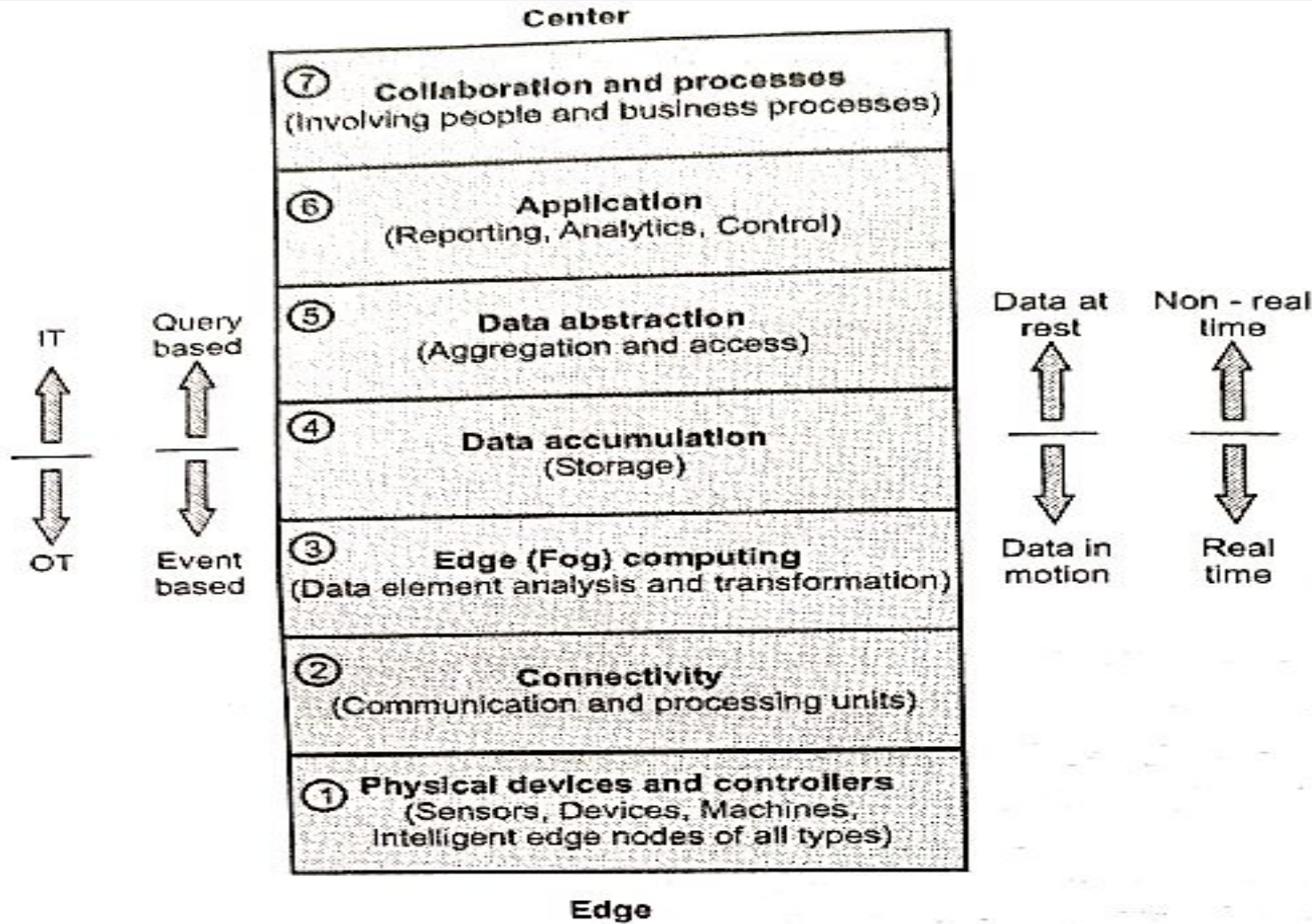
3. Industrial Automation:

Manufacturing plants use M2M communication between machines such as robots, conveyor systems, and quality control sensors. Machines exchange data to optimize production processes, detect faults, and adjust operations automatically, enhancing productivity and reducing downtime.

END TO END ARCHITECTURE

- **Definition:** End-to-End architecture emphasizes seamless communication across all layers of the IoT ecosystem, from devices at the edge to cloud services and applications.
- **Characteristics:**
 1. **Centralized or Distributed:** Can be centralized (where data flows through a central hub or server) or distributed (where data flows between multiple nodes or tiers).
 2. **Holistic View:** Ensures data integrity, security, and management across the entire IoT network.
 3. **Complexity:** Often more complex to implement due to the need for comprehensive integration and management of various components.

END TO END ARCHITECTURE



- End-to-End IoT systems encompass a broader ecosystem involving interconnected devices, networks, and applications, enabling comprehensive data collection, analysis, and decision-making across various domains such as smart cities, healthcare, and logistics.

Fig.: End to End Architecture / IoT WF standardized

FEW EXAMPLES OF END TO END ARCHITECTURE:



1. Smart Cities:

Integrated smart city solutions involve End-to-End IoT architecture where data from various sensors (traffic, weather, pollution) across the city is collected, processed, and analyzed centrally. City administrators use this data to make informed decisions on traffic management, waste management, and resource allocation.

2. Healthcare Monitoring:

Wearable health devices (like fitness trackers or medical monitors) collect vital signs and patient data. This data is transmitted securely through various networks to cloud-based healthcare platforms. Healthcare providers access and analyze this data to monitor patient health remotely, enabling proactive care management.

3. Supply Chain Management:

End-to-End IoT solutions in supply chain management involve tracking goods throughout their journey from manufacturer to retailer or consumer. IoT-enabled tags and sensors on packages communicate location, temperature, and condition data through multiple nodes (e.g., warehouses, transport vehicles) to ensure timely delivery and quality control.

ADVANTAGES OF IOTS

1. Improved customer engagement and communication.
2. Support for technology optimization.
3. Support wide range of data collection.
4. Reduced waste

DISADVANTAGES OF IOTS

1. **Loss of privacy and security:** As all the household appliances, industrial machinery, public sector services like water supply & transport, many other devices all are connected to the internet, a lot information is available on it. This information is prone to attack by hackers.
2. **Flexibility:** Many are concerned about the flexibility of an IoT system to integrate easily with another.
3. **Complexity:** The IoT is a diverse & complex network. Any failure or bugs in the software or hardware will have serious consequences. Even power failure can cause lot of inconvenience.
4. **Compatibility:** Currently, there is no international standard of compatibility for the tagging & monitoring equipment.

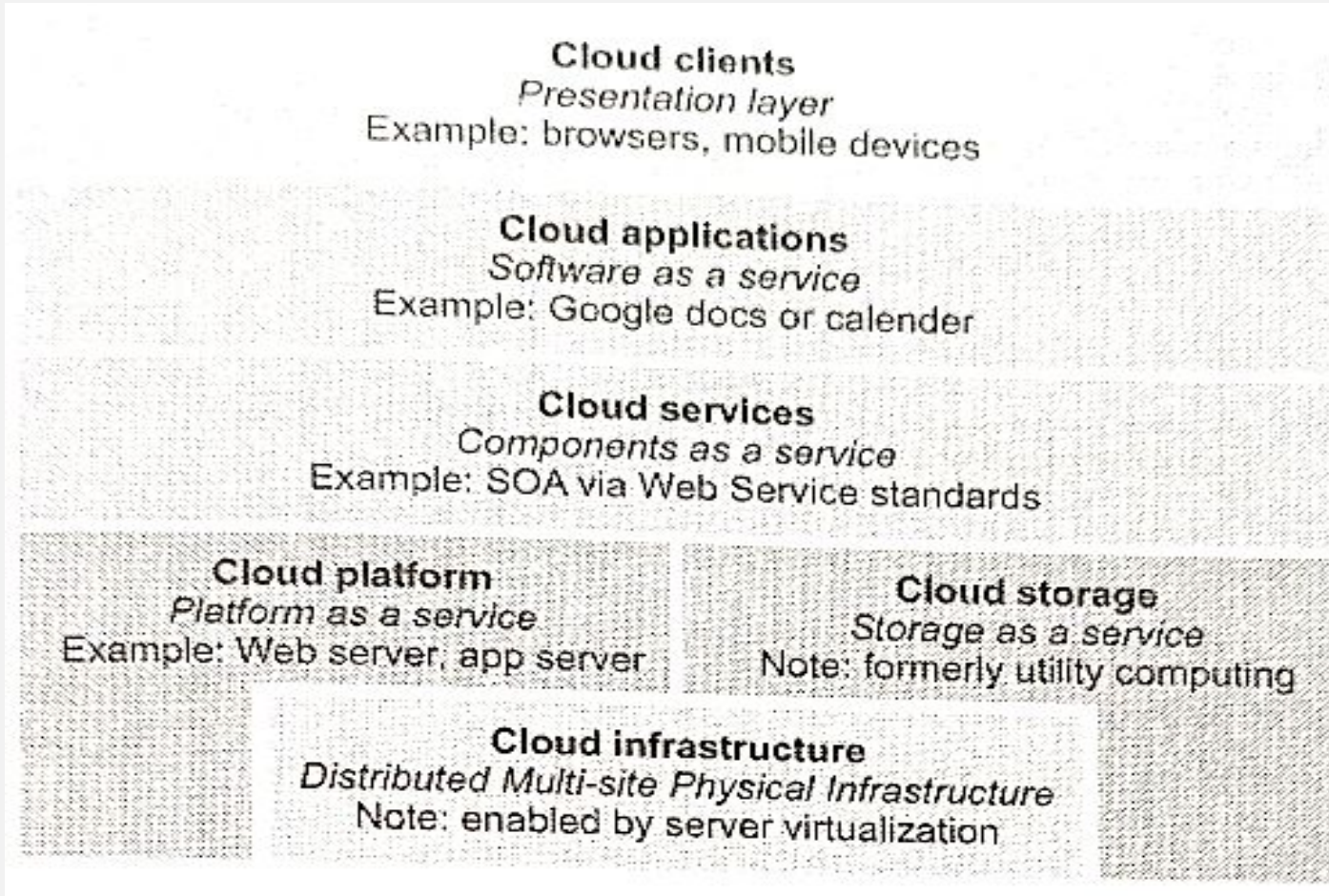
INTERDEPENDENCIES BETWEEN IOT & CLOUD COMPUTING

Cloud computing is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned & released with minimal management effort or service provider interaction.

The cloud paradigm can be delivered using essentially three different service models

- i. Infrastructure as a Service (IaaS)
- ii. Platform as a Service (PaaS)
- iii. Software as a Service (SaaS)

INTERDEPENDENCIES OF IOT & CLOUD COMPUTING



- A Cloud based IoT platform is then a dynamic & flexible resource sharing platform delivering IoT services. It offers scalable resources & services management.
- The exploitation of IoT data depends on massive resources, which should be available when needed and scaled back when not needed.
- Cloud technologies offer decentralized and scalable information processing & analytics and data management capabilities.

Fig: Cloud computing represented as a stack of service offering categories

SUMMARY

- The Internet of Things (IoT) is a network of interconnected devices that share data over the internet. It has evolved from simple machine-to-machine communication to a complex ecosystem of connected devices.
- IoT systems consist of physical objects embedded with sensors and connectivity, allowing them to collect and exchange data for informed decision-making and process automation. Key characteristics of IoT include connectivity, interoperability, scalability, and real-time data processing.
- M2M communication plays a vital role in IoT, enabling seamless automation and efficient data exchange.
- The end-to-end IoT architecture includes layers such as perception, network, and application, facilitating data collection, transmission, analysis, and action triggering.

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Thank You